Predicting Problematic Internet Use

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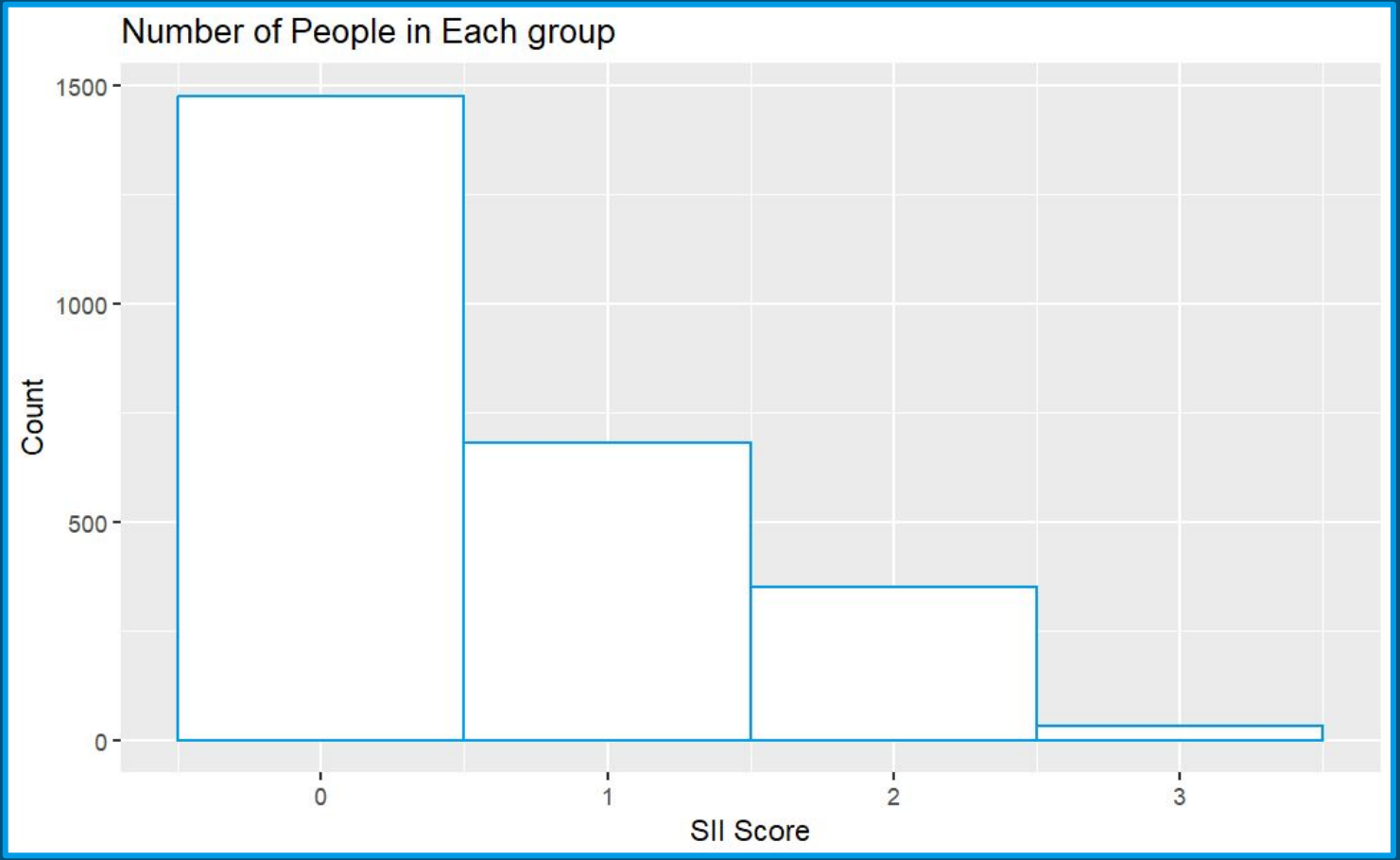
Premise: Kaggle Competition

- Child Mind Institute: Problematic Internet Use
- Main goal of competition is to predict Severity Impairment Index (SII) score
- We also wanted to try and predict Sleep Disturbance Scale (SDS) score, which seems interesting

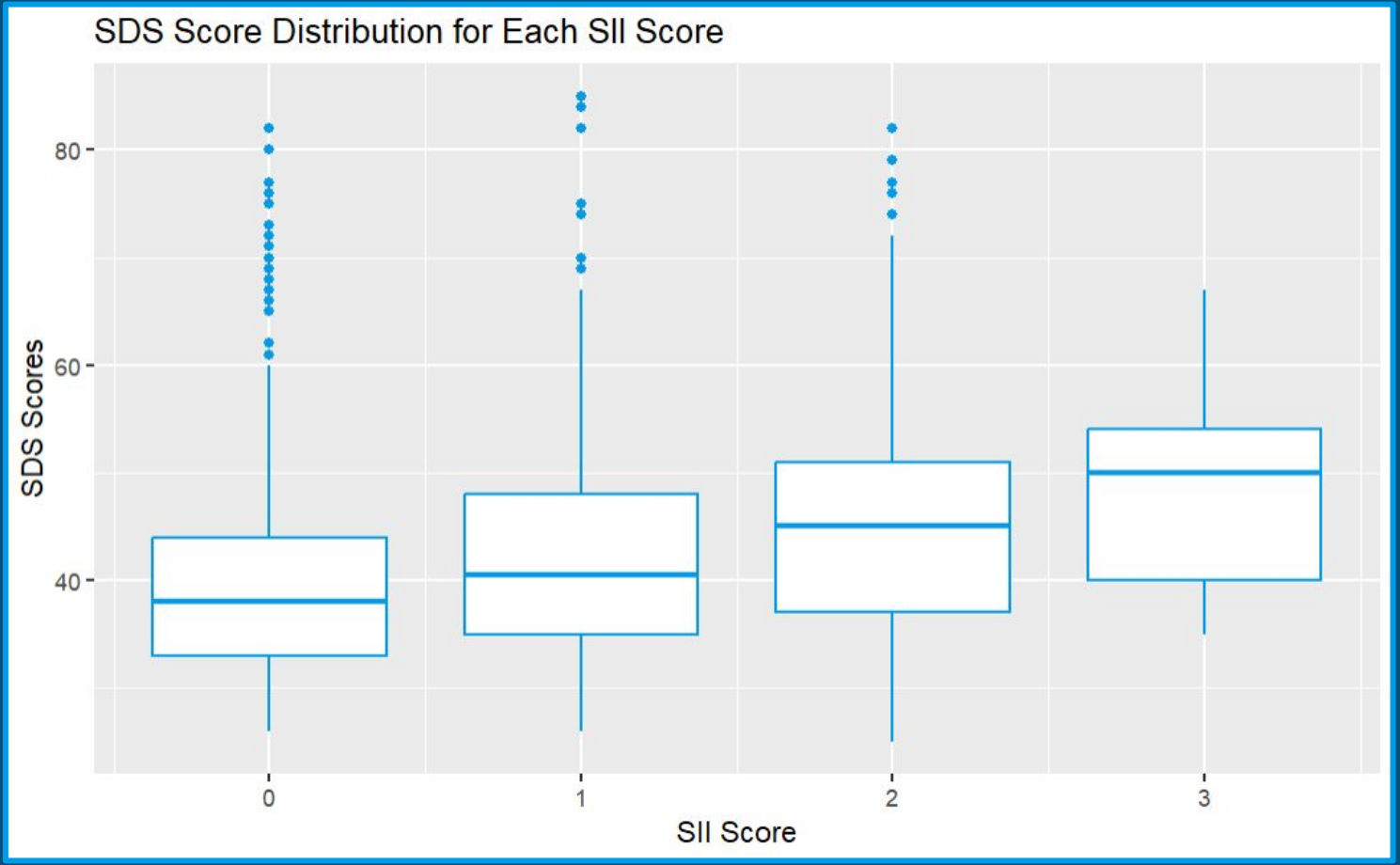
EDA

"SII" stands for "Severity Impairment Index"



EDA

“SDS” stands for “Sleep Disturbance Scale”



Linear Regression

Dealing With NAs

- Removed variables with proportion of NAs over 0.5
- 10 Variables removed this way
- Then removed all observations with proportion of NAs over 0.5
- 1097 observations removed this way

Variable	Total_NAs	NA_Proportion
Physical.Waist_Circumference	3062	0.773
Fitness_Endurance.Max_Stage	3217	0.812
Fitness_Endurance.Time_Mins	3220	0.813
Fitness_Endurance.Time_Sec	3220	0.813
FGC.FGC_GSND	2886	0.729
FGC.FGC_GSND_Zone	2898	0.732
FGC.FGC_GSD	2886	0.729
FGC.FGC_GSD_Zone	2897	0.732
PAQ_A.PAQ_A_Total	3485	0.880
PAQ_C.PAQ_C_Total	2239	0.565

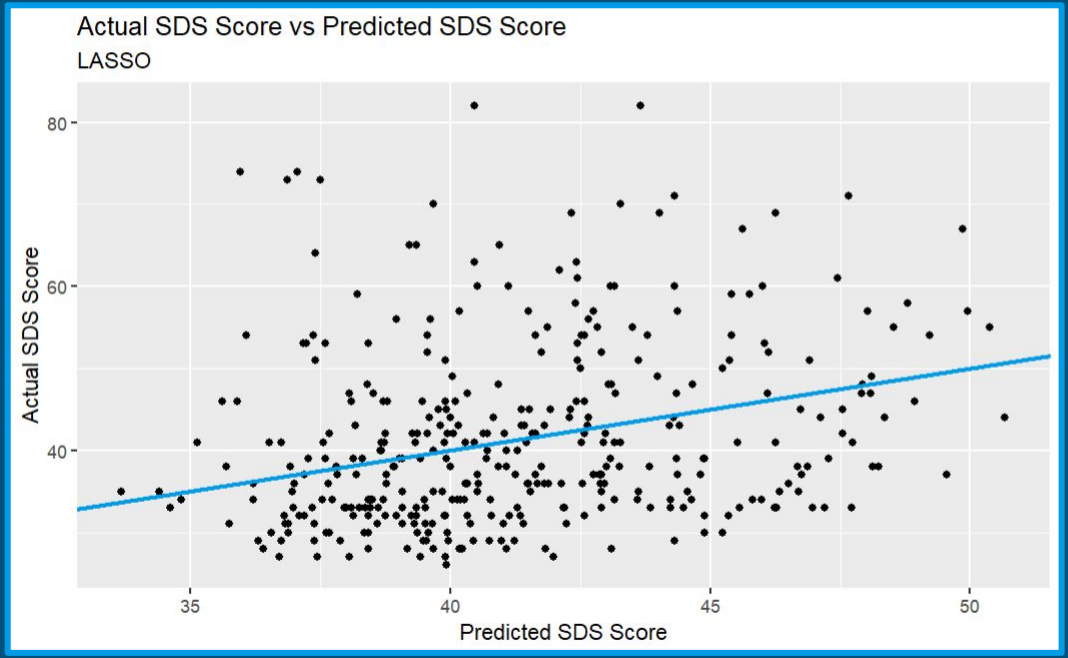
ID	Total_NAs	NA_Proportion
0016bb22	58	0.951
0083e397	58	0.951
0087dd65	58	0.951
0194c251	58	0.951
023e1b6a	58	0.951
040d103f	58	0.951
04ccb838	58	0.951
05dc60ef	58	0.951
0677d22d	58	0.951
0882b065	58	0.951

Dealing With NAs

- Used imputation for rest of NAs
  - mice and tidyr packages were good for this
- There were still some NAs in the data after imputing
- All remaining NAs were in same columns/rows
- Chose to get rid of observations rather than removing more variables
- Final dataset had 61 variables
- Training set had 1584 observations
- Test set had 360 observations

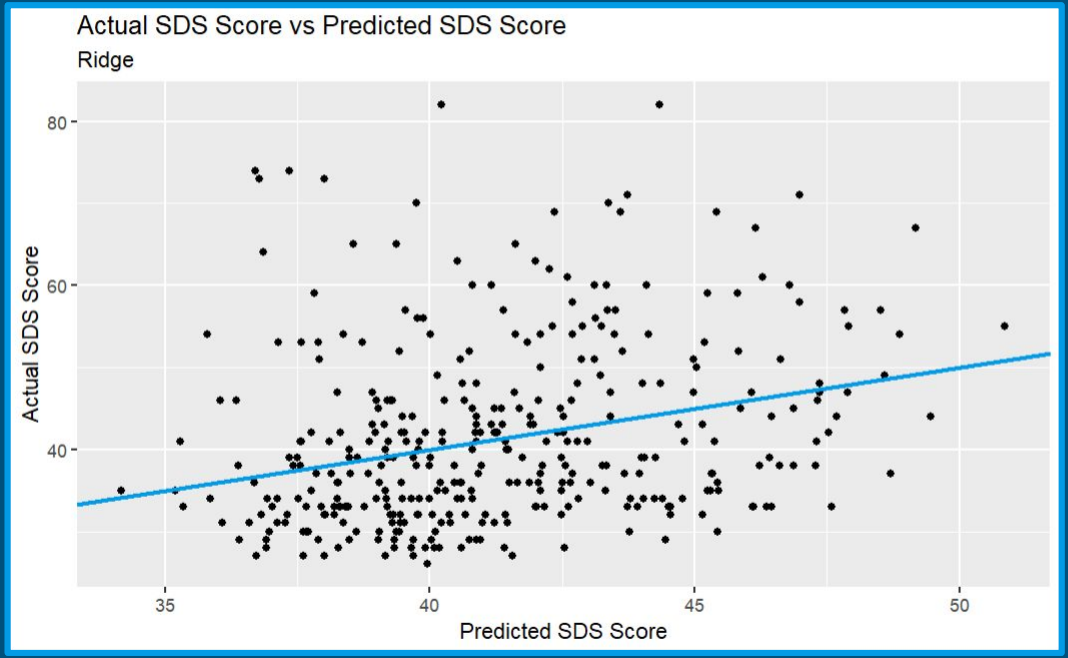
LASSO

- Trying to predict SDS Score
- 28 predictors were used
  - SII score was used
- Only 2 of the 17 Bioelectric Impedance Analysis variables were used
  - BMI and Activity Level
- Test RMSE for the best Lasso model was 10.73954



Ridge

- 28 predictors used again
- Same predictors were used in both Ridge and Lasso
- Different coefficient estimates
- Test RMSE for best Ridge model was 10.6742



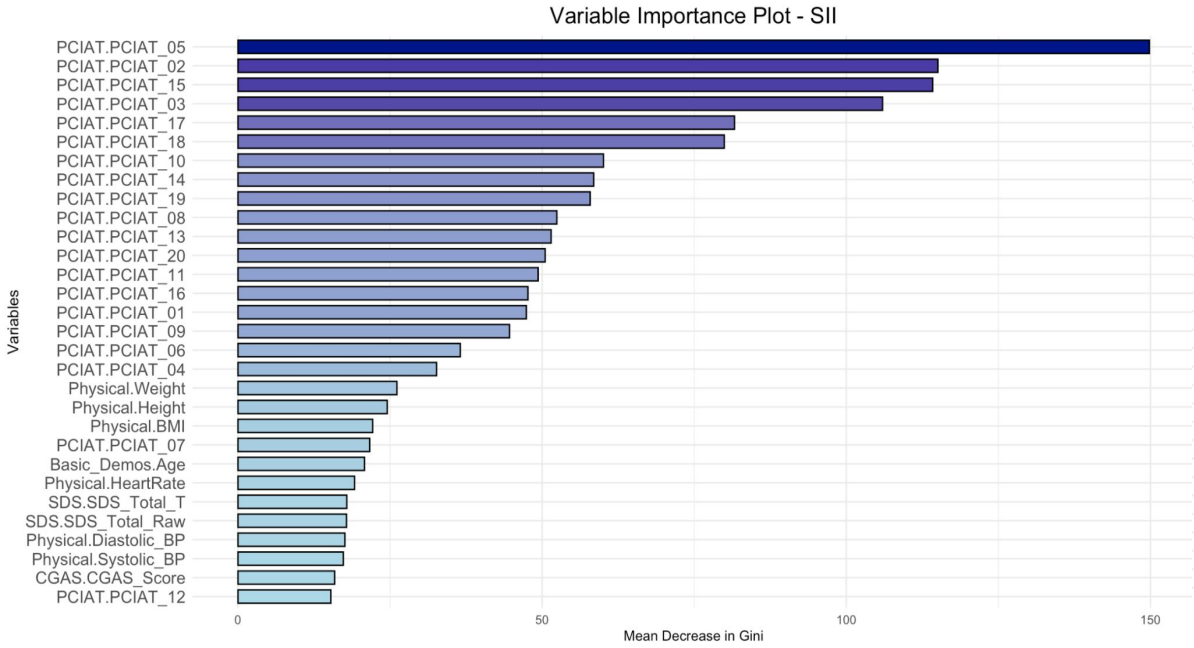
Random Forest

Random Forest

- Data Cleaning:
  - Removed predictors and observations with more than 40% missing data.
  - A total of 36 variables and 1,225 observations were excluded.
- Applied Random Forest to identify key variables associated with:
  - Severity Impairment Index (SII)
  - Sleep Disturbance Scale (SDS)
- **Variable Adjustments**:
  - Excluded the Parent-Child Internet Addiction Test (PCIAT) total score because it is a linear combination of other PCIAT variables.
  - PCIAT variables are categorical, ranging from 0 (strongly disagree) to 5 (strongly agree).

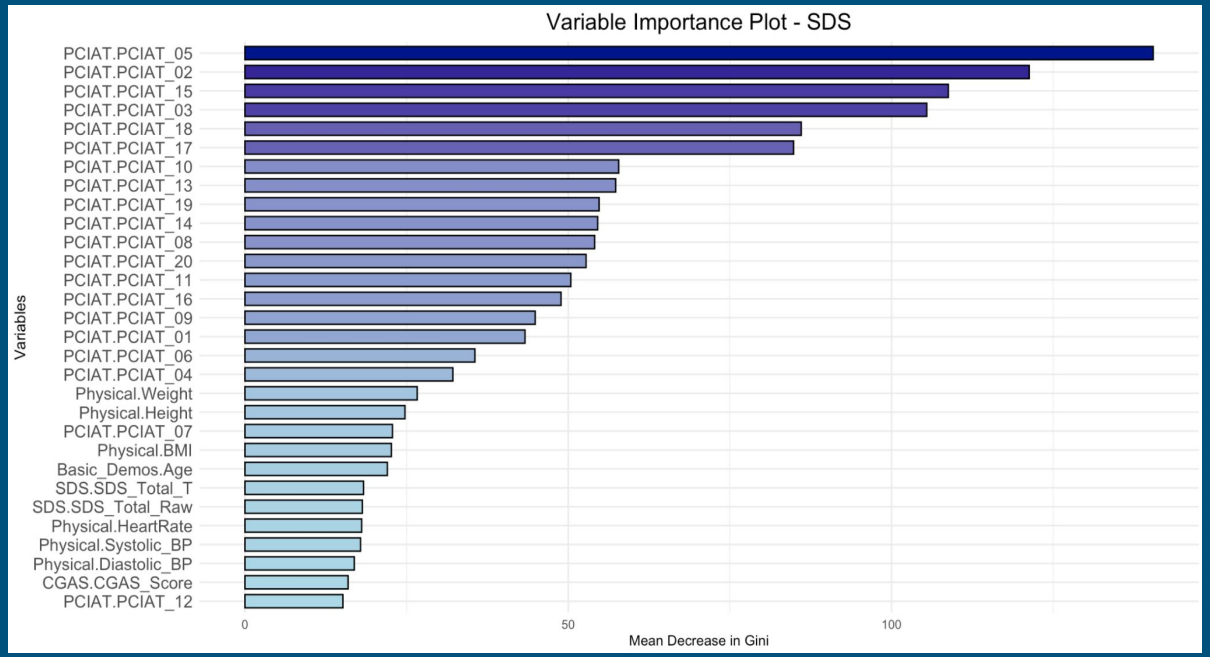
Random Forest

- The plot highlights the most important variables related to the SII, ranked by their Mean Decrease in Gini score
- PCIAT variables are the most strongly associated with SII, with higher scores linked greeted severity.
- Physical measures (Weight, Height, BMI) are the next most important contributors



Random Forest

- This plot reflecting their importance in predicting the Sleep Disturbance Scale (SDS).
- PCIAT variables again show the strongest association with SDS,.
- Physical measures are the second most important contributors, consistent with the SII results.



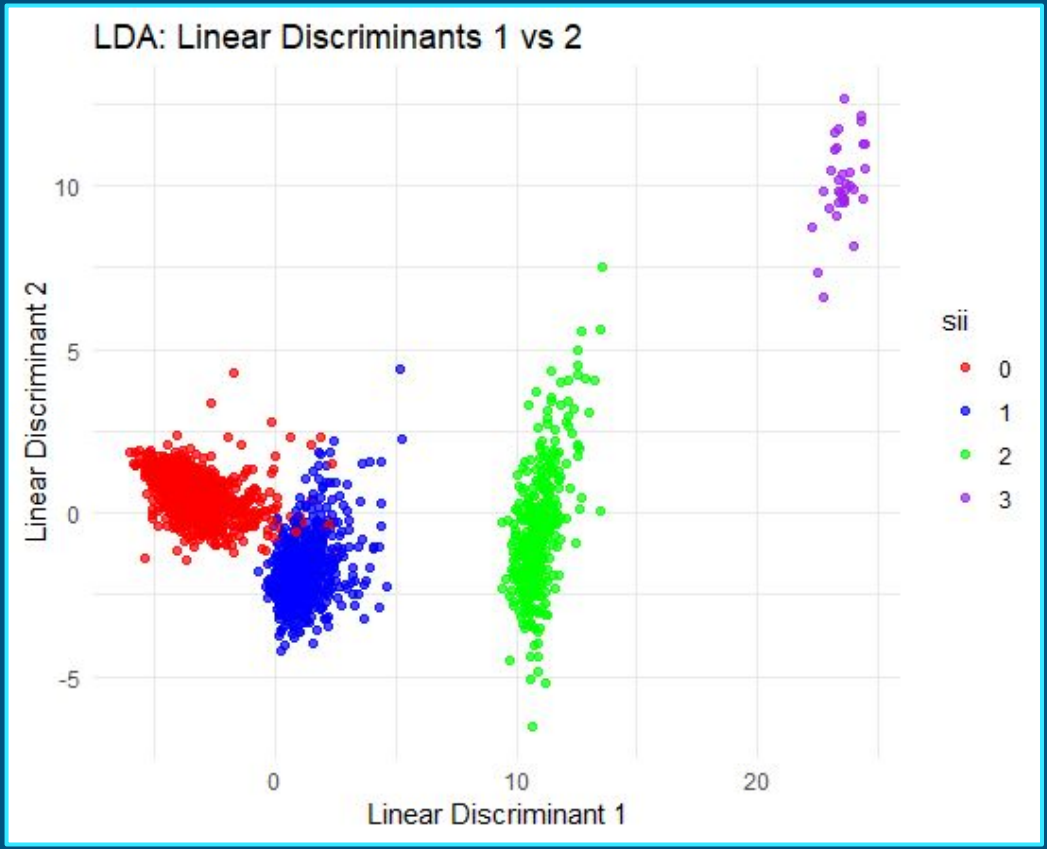
Linear Discriminant Analysis

Dealing with NAs

- Lots of NAs
- Options:
  - **A. Imputation**
    - Median for numerical variables
    - “Missing” for categorical variables
  - **B. Remove rows where response = NA**
    - Followed with imputation
  - **C. Removing Variables**

2	3	4	5	6	7	8	9	10
Fall	5	0	Winter	51	Fall	16.87732	46.00	50.8
Summer	9	0	NA	NA	Fall	14.03559	48.00	46.0
Summer	10	1	Fall	71	Fall	16.64870	56.50	75.6
Winter	9	0	Fall	71	Summer	18.29235	56.00	81.6
Spring	18	1	Summer	NA	NA	NA	NA	NA
Spring	13	1	Winter	50	Summer	22.27995	59.50	112.2
Fall	10	0	NA	NA	Fall	19.66076	55.00	84.6
Fall	10	1	NA	NA	Fall	16.86129	59.25	84.2
Summer	15	0	NA	NA	Spring	NA	NA	NA
Summer	19	1	Summer	NA	NA	NA	NA	NA
Spring	11	1	NA	NA	NA	NA	NA	NA
Fall	11	0	Summer	66	NA	NA	NA	NA
Fall	13	0	NA	NA	Winter	21.07907	57.75	100.0
Spring	12	0	NA	NA	Spring	15.54411	60.00	79.6
Spring	12	0	NA	NA	NA	NA	NA	NA
Winter	7	0	Summer	51	Spring	29.31577	54.00	121.6
Spring	5	1	Summer	80	Spring	17.28450	44.00	47.6
Fall	10	1	NA	NA	Fall	19.89316	55.00	85.6
Winter	6	0	Spring	60	Winter	30.09465	37.50	60.2
Winter	10	0	NA	NA	NA	NA	NA	NA
Winter	14	0	Spring	68	Spring	17.16864	66.50	108.0
Summer	16	0	NA	NA	Summer	29.28586	71.00	210.0
Fall	12	0	Winter	58	Fall	34.18728	60.50	178.0
Winter	13	0	Summer	72	Winter	20.61306	69.00	139.6
Summer	15	0	Fall	70	Summer	21.17778	62.00	115.8

Linear Discriminant Analysis

	0	1	2	3
0	314	4	0	0
1	0	146	0	0
2	0	0	75	0
3	0	0	0	6

Test Data Confusion Matrix: 99.27% Accuracy